

AWS checklist and maintenance guide

Station name	TAS-A	
People present	MCIT, MOLA	instructed by: GOD!
Purpose	<u>standard maintenance</u> / installation / take-down / quick visit / emergency	
Date		
Transport	<u>helicopter</u> / fixed wing / snow mobile / dog sleds / on foot from:	
Weather	<u>warm / cold</u> / very cold / <u>no wind</u> / some wind / rain / cloud cover %: 90	

Notes: Arrived Friday Aug 1. ~1600 Landed, cooked and went to sleep. Woke up 0700 and began work after breakfast. Will camp again here tonight and then be picked up by helo to do TAS-U and TAS-L tomorrow.

Metadata before maintenance (if establishing a new AWS: skip this page)

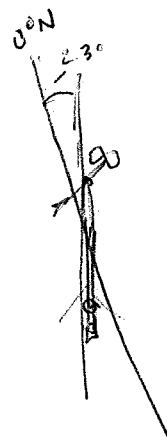
AWS
leads GMT

Time difference between logger clock and UTC	10 s $\oplus$ / -
Adjust logger time to UTC (be sure PC runs on UTC time)	☒ Y / ☐ N
Download data to PC and/or change CF Card (wait for green light)	☒ Y / ☐ N
Name of logger program	2013c
Photo or screen dump of values in fast scan mode	☒ Y / ☐ N
Latitude (dd mm.mmm)	65° 46.824' N
Longitude (dd mm.mmm)	38° 54.129' W
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y / ☐ N
Mast tilt in boom direction (+ if radiometer tilting down)	- 0.8 °
Mast tilt across boom looking from radiometer (- if clockwise)	- 3.7 °
Boom direction relative to north	23 ° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.	~ 2° → W ☒ Y / ☐ N
Radiometer aligned with mast and inclinometer? If not, measure.	☒ Y / ☐ N
Temperature / humidity height (bottom of casing – surface)	255 cm
Sonic ranger on AWS height (membrane – surface)	260 cm
Wind sensor height (center of propeller – surface)	303 cm
Sonic ranger on stakes height (membrane – surface)	99 cm
Free length of stake – outer, holding boom with sonic ranger	115 cm
Free length of stake – middle	130 cm
Free length of stake – outer, holding boom without sonic ranger	123 cm
Free length of ablation hose on/above the ice surface	12 m 83 cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	127 cm
Vertical difference in surface height (at station – at hose) (approx)	30 cm
Length of thermistor string on surface from surface marking	23 cm
Height of surface irregularities	20-30 cm
Stake assembly still standing?	☒ Y / ☐ N / more or less
Apparent damage to sensors? If yes, which?	☐ Y ☒ N

Notes:

station looks good. Little apparent melt since 2013 (when it was established).

from surf. to middle of T-piece in box.



T185-1X

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	080024	SAME	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	01355	→	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna	34	→	
Wind sensor	119282	→	- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly	10/2012	→	
T/hum assembly casing	MP02: 60998574 MERS: 60998579	→	- Fan spins and sounds OK: Y/N
T/hum assembly bracket	4 ?	→	
Temperature probe in T/hum assembly	4 ?	→	
HygroClip in T/hum assembly	60956910	61180280	- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS	4110	→	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel	T2100624 9B1FC52	→	- Output OK: Y/N - Clean: Y/N
GPS antenna	600005	→	- Stuck to top of logger enclosure: Y/N
Data logger	22106/E3200	→	- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader	8525	→	- LED green of flashing orange (active): Y/N
Multiplexer	E5718	→	- Clicking as it should: Y/N
Iridium modem IMEI number	300234011037970	→	- Transmissions tested: Y/N - Problems: Y/N
Barometer	3696480	→	
Sonic ranger on stakes	4284	→	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string	B04	→	- Old string left in ice: Y/N/cut
Ablation hose assembly	5013ABS	→	- Old hose left in ice: Y/N/cut - Air in hose: Y/N removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N - Magnetiz		

TKS-A

356° TRUE

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	(Y) N
Name of logger program	— 2014d
Photo or screen dump of values in fast scan mode	Y/N
Latitude (dd mm.mmm)	SAME
Longitude (dd mm.mmm)	SAME
Photos of tripod, sensors, logger box wiring etc. (more is better)	(Y) N
Mast tilt in boom direction (+ if radiometer tilting down)	+ 1 °
Mast tilt across boom looking from radiometer (- if clockwise)	+ 2.2 °
Boom direction relative to north (radiometer should point to true south)	22° true/magn
Box on wind sensor exactly towards mast? If not, change.	(Y) N
Radiometer aligned with mast and inclinometer? If not, align.	(Y) N
Temperature / humidity height (bottom of casing – surface)	251 cm
Sonic ranger on AWS height (membrane – surface)	260 cm
Wind sensor height (center of propeller – surface)	302 cm
Sonic ranger on stakes height (membrane – surface)	97 cm
Free length of stake – outer, holding boom with sonic ranger	SAME cm
Free length of stake – middle	SAME cm
Free length of stake – outer, holding boom without sonic ranger	SAME cm
Free length of ablation hose on/above the ice surface	SAME cm
Alt: depth of ablation hose using markings on hose	SAME cm
Height of ablation hose fluid level above the ice surface	SAME 122 cm
Vertical difference in surface height (at station – at hose) (approx)	SAME cm
Length of thermistor string on surface from surface marking	SAME cm
Height of surface irregularities	SAME cm

Notes:

Tiltmeter is broken – we do not have a spare.

diff. betw.
inclinom. and
radiom < 0.5°